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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (ISE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21IS602 – INDUSTRIAL ENGINEERING AND MANAGEMENT

20HU701 / 17HU701 – ENGINEERING MANAGEMENT

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) Define management. Elaborate the terms efficiency and effectiveness with examples. 06 L*2 1 1
- b) Explain different management functions. 10 L2 2 2
- c) Discuss Delphi method of forecasting. 04 L2 1 1
2. a) If sales for years 2023, 2022, 2021, 2020 (n=4) were 1700, 1400, 1500 and 1200 respectively. Forecast the sales for 2024 using (i) simple moving average and (ii) weighted moving average with weights 0.4, 0.25, 0.2 and 0.15 for the most recent past years. 06 L3 1 1
- b) Fit a regression line for the following data. Forecast the sales for the year 2024.

Advertisement	Sales
2019	\$1000
2020	\$1300
2021	\$1800
2022	\$1200
2023	\$1380

08 L3 1 1
06 L2 2 2

- c) Differentiate rules and procedures.
3. a) Discuss different types of organization based on internal structure with their advantages and disadvantages. 08 L2 2 2
- b) Differentiate autocratic and democratic leaderships with respect to their merits and demerits. 06 L2 2 2
- c) Explain the different steps in controlling. 06 L2 2 2

Unit – II

4. a) Explain (i) Maslow's Need Hierarchy Theory of motivation and (ii) Herzberg's Two factor theory. 10 L2 3 11
- b) Write a note on X and Y motivational theories. 04 L2 3 11
- c) Explain the process of communication. 06 L2 3 10
5. a) What is a project? Explain the attributes of project with example. 06 L2 4 11

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- b) Draw the Gantt Chart for the following house project.

Task	Follows task(s)	Duration (weeks)	Task Description	Manning Level
A	Start	1.0	Clear site	3
B	Start	0.6	Obtain basic materials	1
C	Start	2.0	Obtain other materials	1
D	B	2.0	Prefabricate wall panels	4
E	B	0.9	Prefabricate roof trusses	3
F	A, B	1.0	Form and pour footings and floor slab	3
G	D, F	0.3	Erect wall panels	4
H	E, G	0.2	Erect roof trusses	4
J	C, H	0.5	Complete roof	3
K	J	2.0	Finish interior	4
L	J	1.0	Finish exterior	2
M	L	0.4	Clean up site	1
N	K, M	0.2	Final inspection and approval	1

- c) What are the advantages and disadvantages of Gantt Chart?

10 L3 4 11
04 L2 4 11

6. a) Define a network with an example. Explain the rules of network construction.

08 L2 4 11

- b) An R&D project has a list of tasks to be performed whose time estimates are given in the following table.

Activity	Follows	Optimistic time	Most likely time	Pessimistic time
A	-	4	6	8
B	-	2	3	10
C	-	6	8	16
D	A	1	2	3
E	B	6	7	8
F	B	6	7	14
G	C, D, E	3	5	7
H	C, D, E	4	11	12
I	F	2	4	6
J	G	2	9	10

- (i) Draw the project network.
(ii) Find the expected time and variance of time for each activity.
(iii) Find the critical path.

12 L3 4 11

Unit – III

7. a) Explain simple interest, compound interest and effective interest rate with relevant formulas for calculating them.

06 L2 5 11

- b) Find the compound interest (CI) on Rs. 12,600 for 2 years at 10% per annum compounded annually.

03 L3 5 11

- c) Calculate the effective annual interest equivalent of nominal rate of 18% per annum compounded half yearly and at that rate what will be the amount received if Rs. 10,000/- is invested for half year?

05 L3 5 11

- c) Explain present worth and future worth with their significance and formula to calculate them.

06 L2 5 11

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|-------|---|----|----|---|----|
| 8. a) | What is depreciation? Explain different types of depreciations with their causes. | 08 | L2 | 5 | 11 |
| b) | If an asset has a first cost of \$50,000 with a \$10,000 estimated salvage value after 5 years, (a) calculate the annual depreciation and (b) calculate the book value of the asset after each year, using straight line depreciation. | 05 | L3 | 5 | 11 |
| c) | Underwater electroacoustic transducers were purchased for use in SONAR applications. The equipment will be DDB depreciated over an expected life of 12 years. There is a first cost of \$25,000 and an estimated salvage of \$2500. Calculate the depreciation and book value for years 1 and 4 using double declining depreciation method. | 07 | L3 | 5 | 11 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
